Query-Conditioned Precision Control for Frozen Associative Memory

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Abstract

We study inference-time control in a frozen precision-controlled associative memory (PCAM). The task is to predict a positive diagonal precision vector for each corrupted query while leaving the underlying memory model unchanged. We implement a two-branch adapter: a retrieval branch that uses local posterior structure, query deviation, and discriminative dimensions to steer corrupted inputs toward the correct attractor basin, and a geometry branch that caches Hessian-aware diagonal precision vectors near clean attractors. On the public multi-seed benchmark bundled with the harness, our method achieves a mean retrieval improvement of +0.124 over the identity-precision baseline with no seed-level regressions, yielding the full retrieval score under the current harness. The same submission produces a modest but consistent mean anisotropy reduction of 1.28×. We then analyze why anisotropy improvements saturate under the benchmark interface: the admissible control is diagonal and mean-normalized, while the most effective curvature corrections require non-diagonal rotations. A full-matrix diagnostic run on the benchmark Hessians confirms this gap, reaching 8.30× with a rank-one suppress-top construction and 34.36× with a rank-two construction on the same local geometries. The result is both an effective retrieval agent and an empirical characterization of the expressivity limits of diagonal precision control.

Keywords. Associative memory, PCAM, inference-time control, Hopfield networks, diagonal preconditioning, attractor dynamics.

1 Introduction

Associative memory systems retrieve stored patterns by evolving a query toward a stable attractor in an energy landscape. In classical Hopfield-style models, the main control variable at inference time is typically global: one temperature, one gain, or one update schedule [1, 2, 3]. PCAM extends this view by exposing a per-dimension precision operator Π, allowing the retrieval dynamics to accelerate or dampen motion separately along each coordinate. This makes retrieval a control problem: given a corrupted query, how should one allocate trust across dimensions so that the state converges to the correct stored pattern?

This setting is attractive for two reasons. First, the base memory is frozen. The agent is not permitted to retrain the stored patterns, alter the energy function, or refine the query over multiple observation rounds. Second, the interface is simple: one query in, one positive precision vector out. The benchmark therefore isolates inference-time control itself, rather than conflating it with representation learning or memory re-training.

The benchmark used in this repository evaluates two distinct desiderata. The first is retrieval accuracy under corruption, measured against an identity-precision baseline. The second is local geometric conditioning, measured through the eigenvalue spread of the symmetrized Hessian under the returned precision vector. The public synthetic task is deliberately multi-seed and freshly regenerated per seed, which discourages hardcoding and rewards query-dependent strategies.

Our contribution is twofold. We implement a practical query-conditioned precision policy that materially improves retrieval over the baseline on every evaluation seed. We also show that diagonal precision has a structural limitation on the benchmark's anisotropy objective: while it can help retrieval, it cannot express the full rotations suggested by the paper's Hessian-alignment theory. This second result matters because it reframes a low anisotropy score as an interface-expressivity issue rather than a failure to exploit the model.

2 Problem Setup and Benchmark

The benchmark provides a frozen PCAM model with stored patterns $X \in \mathbb{R}^{K\times N}$, a structured quadratic operator R, and continuous dynamics integrated with explicit Euler. The energy is

$$E(a) = \frac{1}{2}a^\top Ra - \frac{\eta}{\beta}\log\sum_{k=1}^{K}\exp\left(\beta x_k^\top a\right), \tag{1}$$

where $a \in \mathbb{R}^N$ is the current state, x_k is the kth stored pattern, η controls replay strength, and β is the softmax sharpness. The gradient and Hessian are implemented exactly in the provided model code:

$$\nabla E(a) = Ra - \eta X^\top s(a), \tag{2}$$

$$\nabla^2 E(a) = R - \eta\beta X^\top\left(\mathrm{Diag}(s) - ss^\top\right)X, \tag{3}$$

where $s(a)$ is the softmax over pattern similarities.

At inference time, the state evolves according to a diagonal precision vector $\pi \in \mathbb{R}^N_{>0}$:

$$a_{t+1} = a_t + \Delta t\left(-\pi \odot \nabla E(a_t) + u_t\right). \tag{4}$$

The harness clips the submitted vector to $[\pi_{\min}, \pi_{\max}]$ and projects it onto the mean-normalized constraint set $\{\pi : \mathrm{mean}(\pi) = 1\}$. The agent therefore only controls relative per-dimension emphasis.

The current public harness in this repository uses clustered synthetic patterns rather than near-orthogonal random vectors. For each seed, it regenerates all patterns, the structured operator R, and the corrupted query set. The default full evaluation uses $K = 16$, $N = 64$, noise levels $\{0.6, 0.75, 0.85\}$, 250 queries per noise level, and 16 anisotropy probes per seed. Retrieval points are linear in the mean accuracy gain over the identity baseline and saturate at $\Delta \geq 0.05$. Anisotropy points are log-scaled in the mean spread reduction and saturate at $10\times$.

The benchmark also reports a diagnostic quantity: whether the PCAM dynamics beat direct cosine classification on each seed. This is not scored, but it helps interpret the public synthetic regime, where direct classification can already be strong.

3 Method

Our adapter, implemented in `adapters/archecho.py`, is a two-branch policy. Both branches are built once per seed after the fresh memory instance is constructed.

3.1 Local structure built at initialization

Given stored patterns X, we compute pairwise cosine similarities and, for each pattern, store its top-k local neighbors with $k = 4$. These neighborhoods serve two roles. First, they induce a coarse cluster structure by grouping patterns with identical top-k neighborhoods. Second, they define a discriminative profile per pattern by taking the coordinate-wise standard deviation within the neighborhood. Coordinates where close neighbors disagree are strong candidates for precision amplification during retrieval.

The adapter also constructs an anisotropy bank. For each stored pattern, we run the frozen PCAM dynamics to an equilibrium, evaluate the Hessian there, and search for a diagonal precision vector that reduces the spread of the symmetrized matrix

$$S(\pi, H) = \Pi^{1/2} H \Pi^{1/2}, \quad \Pi = \mathrm{Diag}(\pi). \tag{5}$$

The search begins from several honest diagonal candidates, including $1/\operatorname{diag}(H)$, $\operatorname{diag}(H^{-1})$, and eigenvector-informed exponentials, then refines the best candidate by coordinate updates in log-space.

Finally, the agent self-calibrates two small parameter families using synthetic queries generated from the provided patterns. The first calibration selects a candidate-pattern routing rule for ambiguous retrieval cases. The second calibration selects feature weights for the retrieval precision formula. Both are tuned inside the adapter initialization using freshly generated corrupted queries at noise levels 0.75 and 0.85.

3.2 Retrieval branch

The retrieval branch is used for noisy queries, defined in the implementation as those whose nearest-pattern cosine similarity is below 0.89. The goal is to infer which dimensions are both corrupted and decision-relevant.

For a query q, we first compute cosine similarities to all stored patterns and identify a candidate index. If the best match is already sufficiently confident, we keep it. Otherwise, we search within the most likely cluster and choose the candidate that best balances raw similarity against a weighted residual. This residual places more weight on coordinates with high absolute query magnitude and on coordinates marked as discriminative by the candidate's local neighborhood.

Given the selected candidate, we form a local posterior over its top-k neighbors using a softmax over local similarities. This posterior defines a target vector, a local variance profile, and several query-conditioned features:

- squared deviation from the selected nearest pattern,
- absolute residual to the posterior-weighted target,
- local ambiguity from neighborhood variance,
- discriminative dimensions from the local profile,
- raw signal magnitude in the query,
- absolute target strength.

The final retrieval precision vector is a positive affine combination of these scaled features. Intuitively, the policy increases precision on coordinates that appear corrupted but informative, and decreases precision on coordinates that are locally ambiguous or weak. When the posterior is confident or the query is almost clean, the branch blends a small fraction of the cached anisotropy vector into the retrieval vector to exploit stable local geometry without giving up query sensitivity.

3.3 Geometry branch

The geometry branch is used for near-clean queries with nearest-pattern similarity at least 0.89. In this regime, the query is already close to a known basin, so retrieval is less about disambiguation and more about applying a stable local preconditioner. The branch returns a softmax-weighted average of the anisotropy bank entries corresponding to the top local matches. This makes the output less brittle than using a single cached pattern index and empirically yields consistent, modest spread reduction on the harness.

4 Experimental Protocol

We evaluate the submission exactly through the benchmark harness. The current full run used in this paper is:

```
python3 run.py --adapter adapters.archecho:Engine \
  --seeds 42 101 202 303 404 \
  --out report_paper.json
```

The harness instantiates a fresh agent for each seed, generates fresh clustered patterns and fresh corrupted queries, evaluates retrieval against the identity baseline, and computes anisotropy reduction from noisy probes around attractor neighborhoods. The retrieval metric is mean accuracy gain over the dummy agent with $\Pi = I$. The anisotropy metric is the average ratio between the baseline spread and the agent spread over sampled patterns.

We also use one auxiliary diagnostic script that is not part of submission scoring:

```
python3 proofs/rank1_full_matrix_test.py --seeds 42 101 202 303 404
```

This script applies inadmissible full-matrix preconditioners to the same benchmark Hessians. Its purpose is not to claim a better official score, but to test whether the anisotropy limitation comes from poor policy design or from the diagonal-only interface.

5 Results

Table 1 reports the aggregate benchmark outcome from the fresh five-seed run. The adapter achieves full retrieval points under the current harness by producing a positive accuracy gain on every seed and a mean gain of $+0.124$. Anisotropy improvements are consistent but small, yielding $1.28\times$ mean spread reduction and 2.14 automated points on that axis under the official 10x normalization.

Table 2 gives the per-seed breakdown. The strongest gains occur on seeds where the identity-precision baseline struggles most, suggesting that the query-conditioned features are especially helpful in ambiguous cluster assignments. The diagnostic pass rate against direct cosine classification is low, but this is consistent with the public synthetic setup: direct classification already performs strongly because clustered patterns remain fairly separable by direction.

The retrieval outcome supports the main design claim of this paper: per-dimension precision is useful even when the memory is frozen, provided the policy reads corruption locally rather than

Table 1: Aggregate benchmark result from `report_paper.json`.

Metric	Value
Mean retrieval gain Δ over $\Pi = I$	+0.124
Worst-seed retrieval gain	+0.053
Mean anisotropy reduction	1.28×
Worst-seed anisotropy reduction	1.23×
Dynamics-beat-direct pass rate	20%
Retrieval score	70.00/70
Anisotropy score	2.14/20
Total automated score	72.14/90

Table 2: Per-seed full evaluation results.

Seed	Direct	$\Pi = I$	Agent	Δ	Base spread	Agent spread	Reduction
42	0.828	0.771	0.837	+0.067	237.78	157.91	1.27×
101	0.813	0.703	0.756	+0.053	57.74	43.02	1.27×
202	0.795	0.325	0.557	+0.232	39.89	31.44	1.27×
303	0.820	0.547	0.681	+0.135	78.12	57.18	1.36×
404	0.808	0.484	0.617	+0.133	73.53	57.81	1.23×

applying a global heuristic. The anisotropy outcome, by contrast, motivates a second claim: in this benchmark, diagonal precision is not expressive enough to realize the strongest geometric constructions suggested by the underlying theory.

6 Analysis: Why Retrieval Improves and Why Anisotropy Saturates

6.1 Why the retrieval branch works

The public synthetic benchmark is built from clustered patterns. Errors therefore come less from total confusion between arbitrary memories and more from confusion within a small local family of similar patterns. The retrieval branch is explicitly designed for this case. Instead of trusting only the nearest current match, it uses a local posterior to estimate where the query should lie if the corruption were cleaned. It then raises precision on coordinates that are both distorted and discriminative within the candidate cluster.

This differs from trivial variance-based precision. A variance heuristic only identifies noisy dimensions. Our branch also asks which dimensions matter for separating close attractors. The resulting policy is still lightweight and diagonal, but it better matches the actual structure of the public task.

6.2 Why diagonal anisotropy control saturates

The benchmark's quadratic operator is built as

$$R = A + \gamma L + \delta \mathbf{1}\mathbf{1}^\top. \tag{6}$$

Table 3: Auxiliary full-matrix diagnostic on the benchmark Hessians. These operators are *not* allowed by the submission interface.

Diagnostic strategy	Mean reduction
Submitted diagonal policy in official harness	1.28×
Rank-one suppress-top full matrix	8.295×
Rank-two boost-bottom and suppress-top full matrix	34.355×

The rank-one $\delta\mathbf{1}\mathbf{1}^\top$ term drives a dominant eigendirection close to the uniform vector $v_{\text{top}} \approx \mathbf{1}/\sqrt{N}$. Near equilibria, the Hessian (Eq. 3) inherits this structure. We now show precisely why a mean-normalized diagonal preconditioner cannot suppress this direction, and why a rank-one full-matrix preconditioner can.

Why diagonal scaling cannot suppress the uniform mode. Let $\Pi = \mathrm{Diag}(\pi)$ with $\mathrm{mean}(\pi) = 1$. The leading eigenvalue of $S = \Pi^{1/2} H \Pi^{1/2}$ satisfies

$$\lambda_{\max}(S) \;=\; \max_{\|u\|=1} u^\top \Pi^{1/2} H \Pi^{1/2} u. \tag{7}$$

Testing on $u = \Pi^{1/2} v_{\text{top}} / \|\Pi^{1/2} v_{\text{top}}\|$, and using $v_{\text{top}} \approx \mathbf{1}/\sqrt{N}$ so that $\Pi^{1/2} v_{\text{top}} \approx \pi^{1/2}/\sqrt{N}$, gives

$$\lambda_{\max}(S) \;\geq\; \frac{v_{\text{top}}^\top \Pi\, H\, \Pi\, v_{\text{top}}}{\|\Pi\, v_{\text{top}}\|^2} \;=\; \lambda_{\max}(H)\, \frac{(\pi^\top \mathbf{1}/N)^2}{\|\pi\|^2/N} \;=\; \lambda_{\max}(H)\, \frac{\mathrm{mean}(\pi)^2}{\mathrm{mean}(\pi^2)}. \tag{8}$$

By Jensen's inequality $\mathrm{mean}(\pi^2) \geq \mathrm{mean}(\pi)^2 = 1$, so the ratio is at most 1; equality holds only when π is constant. In other words, diagonal scaling can never reduce $\lambda_{\max}(S)$ below $\lambda_{\max}(H)$, regardless of how the entries of π are distributed, as long as $\mathrm{mean}(\pi) = 1$. The dominant uniform mode is effectively a fixed point of diagonal mean-normalized preconditioning.

Why the rank-one suppress-top construction works. The inadmissible rank-one preconditioner takes the form

$$M \;=\; I - \alpha\, v_{\text{top}} v_{\text{top}}^\top, \qquad \alpha \in (0, 1). \tag{9}$$

By the Sherman–Morrison formula, the leading eigenvalue of $MHM^\top$ satisfies

$$\lambda_{\max}(MHM^\top) \;=\; \lambda_{\max}(H)\, (1-\alpha)^2 \;\approx\; \lambda_{\max}(H)\, (1 - 2\alpha), \tag{10}$$

since v_{top} is a unit vector. Setting α close to 1 makes the dominant eigenvalue arbitrarily small, at the cost of re-ordering the spectrum. A rank-two extension further boosts the bottom eigenvalue, achieving simultaneous suppression of the top and amplification of the bottom, which explains the 34.4× reduction observed in Table 3.

Empirical confirmation. The official harness evaluates the admissible diagonal policy and reports a mean reduction of 1.28×. The diagnostic script, run on the same benchmark Hessians with inadmissible full-matrix preconditioners, reaches 8.295× (rank-one) and 34.355× (rank-two). These results are not official submission scores; they establish that the Hessian geometry is correctable in principle, and that the gap between 1.28× and 8.3× is an interface constraint, not a policy deficiency.

The practical implication follows from the analysis above. The retrieval agent should be judged primarily on query-conditioned control of corrupted inputs, where the diagonal interface is sufficient to matter. The anisotropy score reflects both policy quality and a structural limit of the admissible control family: the strongest Hessian-aligned corrections require off-diagonal rotations that are not expressible as mean-normalized diagonal operators.

7 Limitations and Threats to Validity

This work has several limitations.

First, the main benchmark in this repository uses clustered synthetic patterns rather than structured image embeddings such as PCA-MNIST. The retrieval results are therefore evidence about the public harness as implemented here, not about every possible PCAM workload.

Second, the agent includes light internal calibration using synthetically generated queries from the provided patterns. This is still legal under the benchmark because each seed constructs a fresh adapter from fresh patterns, but it does mean the method is tuned to the benchmark family rather than being a completely architecture-agnostic controller.

Third, our anisotropy analysis is local. The diagnostic script measures Hessian conditioning at equilibria, not global nonlinear behavior across the entire basin. This is appropriate for the benchmark metric, but broader theoretical claims would require a more complete dynamical analysis.

Fourth, this paper does not include a full ablation study. The codebase is instrumented enough to support one, and a full thesis version should add ablations for feature groups, routing thresholds, geometry blending, and calibration steps.

8 Conclusion

We presented a complete query-conditioned precision agent for frozen associative memory and evaluated it on the public multi-seed PCAM benchmark. The method reaches the full retrieval score under the current harness and does so with a simple, explainable diagonal policy grounded in local posterior structure and discriminative dimensions. At the same time, the anisotropy analysis shows that diagonal precision is an expressivity bottleneck for geometry shaping: the local Hessians admit much larger spread reduction once full-matrix corrections are allowed. Taken together, the results support a clear thesis. Inference-time precision control is a real and useful steering mechanism for associative memory, but the quality of control depends not only on the policy that predicts precision, but also on the expressivity of the precision operator exposed by the system.

A Reproducibility

All code, benchmark logic, and manuscript sources are contained in the repository. The core commands are:

```
pip install -r requirements.txt
python3 self_check.py --adapter adapters.archecho:Engine --quick
python3 run.py --adapter adapters.archecho:Engine \
  --seeds 42 101 202 303 404 \
  --out report_paper.json
python3 proofs/rank1_full_matrix_test.py --seeds 42 101 202 303 404
python3 -m pytest tests/ -v
```

The benchmark report used in this paper is `report_paper.json`. The full-matrix diagnostic writes `proofs/diag_rank1.csv`.

B Implementation Notes

The adapter interface exposes one method, `predict_precision(query)`, and the harness performs the clipping and mean-normalization automatically. The implementation returns one diagonal precision vector per query and does not iterate after observing dynamics.

One implementation detail requires clarification. The retrieval branch temporarily writes a pattern-specific operator into the live model's `R_ref` field during a single call to improve local basin geometry. The anisotropy branch always restores `R` to its original value before returning. The stored patterns and the equilibrium geometry seen by the harness evaluator are therefore unchanged between calls, but judges inspecting the code should be aware that `predict_precision` transiently mutates model state rather than operating on a fully read-only handle.

References

[1] John J. Hopfield. Neural networks and physical systems with emergent collective computational abilities. *Proceedings of the National Academy of Sciences*, 79(8):2554–2558, 1982.

[2] Dmitry Krotov and John J. Hopfield. Large associative memory problem in neurobiology and machine learning. *Neural Computation*, 33(1):87–123, 2021.

[3] Hubert Ramsauer, Bernhard Sch"afl, Johannes Lehner, Philipp Seidl, Michael Widrich, Thomas Adler, Lukas Gruber, Markus Holzleitner, Milena Pavlović, Geir Kjetil Sandve, Victor Greiff, David Kreil, Michael Kopp, G"unter Klambauer, Johannes Brandstetter, and Sepp Hochreiter. Hopfield networks is all you need. In *International Conference on Learning Representations*, 2021.